SMITHIAN FOLD THEORY - MATHEMATICS BRANCH PAPER 001

From Fold to Mathematics

An Exact, Parameter-Free and Machine-Closed Derivation of Mathematical Foundations from Smithian Fold Theory

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Third clean-room reconstruction - Mathematics current-knowledge inventory complete
Twelve admitted derivations - 7,424 generated candidate structures
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Abstract

This paper reports the completed Mathematics branch of the third clean-room reconstruction of Smithian Fold Theory (SFT). Starting only from the ten model-admitted Foundation receipts, it derives exact arithmetic and number structure; discrete mathematics; combinatorics; graph and network theory; algebraic structures; order and lattice structure; finite computational geometry and topology; exact probability and statistics; optimization; finite dynamical systems; logic and proof theory; and category, type and compositional structure. No conventional mathematical axiom system, semantic numerical zero, negative quantity, irrational or imaginary proof value, floating-point proof quantity, completed infinity, ungenerated continuum, stochastic cause, fitted parameter, pretrained model or application result is admitted as a premise.

The twelve claim grammars execute 7,424 generated candidate structures. Every generated candidate receives an exact decision; each grammar has exactly one all-preserving survivor. Every claim passes minimality, named-shape uniqueness, a depth-independent base/successor certificate, false-premise, source-tamper, artifact-tamper and boundary controls, cryptographic sealing and implementation-distinct recomputation. The branch therefore contains twelve depth-independently closed, model-admitted and independently replicated engine receipts. A complete repository verification currently passes 111 unit and end-to-end tests, observes 1,264 of 1,264 executable core-engine lines and independently reruns all 22 admitted Foundation and Mathematics derivations in dependency order.

The contribution is not a relabelling of standard mathematics. Familiar terms enter only after each SFT derivation seals, as correspondence vocabulary. The primary objects are complete generated traces, exact held/whole parts, pair cells, held labels, canonical finite forms, relations and proof traces. Empty and identity cases use structural empty One rather than numerical zero; direction and opposition use held orientation rather than signed magnitude; uncertainty is a relation between deterministic support and observation rather than an imported random cause. Every prose claim is linked to machine-readable census, elimination, control, certificate and receipt evidence.

Keywords: Smithian Fold Theory; foundations of mathematics; exact arithmetic; discrete mathematics; combinatorics; graph theory; algebra; lattice theory; finite topology; probability; optimization; dynamical systems; proof theory; category theory; open computational science.

1. Central scientific claim

The exact claim of this paper is:

> *Within the frozen SFT V3 Mathematics current-knowledge inventory, every one of the twelve registered mathematical-foundation obligations has a depth-independent, model-admitted and independently replicated engine receipt; the inventory contains no unclassified or frontier obligation.*

The inventory identity is `sha256:d8eaad7d40d75bb293ee49b1c6401aeb7a617068e597b2cc4eeb00a7879fc3d3`. It fixes the branch boundary and claim order before the paper is evaluated. Branch closure means closure of these twelve general generated-finite mathematical kernels. It does not assert a completed infinite universe or permit a familiar named structure to inherit properties without its own generated witness. Applications in computation, physics, biology, engineering, Fold Protein, Fold Chess, Fold Go and Unison AI did not select any law in this paper.

2. Standalone dependency foundation

This paper is standalone in exposition but not dependency-free in the scientific census. The prior Foundation branch supplies ten exact receipts: operational occurrence, structural One, complete positive finite count, exact held/whole part coordinates, the minimal Fold, cross-partition part equivalence, Fold assembly, finite form grammar, canonical form identity and the one-way measurement boundary. Those are cited as admitted dependencies, not copied assumptions. The published Foundation paper remains a separate work and is not modified by this paper.

The sole root theorem remains *there is no nothing*. Every dependency path terminates at its admitted receipt. Each Mathematics registration contains empty axiom and free-parameter tuples. Prior SFT versions, conventional theorems and correspondence names are excluded from derivational source manifests.

3. Exact mathematical constitution

The admitted domain contains structural One, complete positive finite generation traces, exact positive held/whole parts, canonical labels, fibres, branches, words, pair cells, relations, forms and proof traces. The domain does not contain semantic numerical zero, a negative quantity, an irrational or imaginary proof value, floating proof arithmetic, a completed infinite set or an ungenerated continuum. An empty selection, identity path or no-transition duration is the structural empty-One form. Opposed direction is a distinct held orientation with its positive remainder trace.

Python host integers, Boolean checks, empty containers and process status codes are implementation mechanics. They do not become SFT proof objects. The scientific sources explicitly return structural forms or exact positive part relations at the boundary where a conventional implementation might otherwise introduce zero, negative or floating values. The admission engine independently rejects registered axioms, free parameters, missing dependencies, incomplete censuses, multiple survivors, failed controls and source drift.

4. What forced, closed and independently validated mean

A declaration is not forced because it is elegant or familiar. Every claim specifies a product grammar of structural questions. Each coordinate supplies at least one explicitly rejected alternative and exactly one all-preserving alternative. The complete Cartesian product is generated in canonical order. Candidate identities, exact forms and trace hashes are recorded. Every candidate receives one survival decision, elimination reason and proof hash. Exactly one candidate must survive.

Closure then requires more than the finite run. Minimality shows that replacing any preserving coordinate loses a registered requirement and that no extra rule remains. Named-shape uniqueness shows that the full product contains only one all-preserving coordinate. A claim-specific structural One base and successor certificate extends the classification to every generated finite depth. This is depth-independent closure over finite generation, not a completed infinite object.

Four controls are mandatory. A false premise must be rejected; a changed official source identity must be detected; a missing, duplicated or additional survivor must fail; and an excluded object or answer-producing model must halt at the boundary. After the derivation is sealed, a separate Python process whose file hash is not part of the scientific implementation regenerates the literal candidate product, decision vector, unique survivor, closure flags and controls. Only the shared admission engine can add the receipt to the census.

5. Dependency order and executed census

Order	Claim	Candidate structures	Dimensions	Closure
1	`SFT-MATH-EXACT-ARITHMETIC-001`	256	8	depth-independent
2	`SFT-MATH-DISCRETE-001`	256	8	depth-independent
3	`SFT-MATH-COMBINATORICS-001`	256	8	depth-independent
4	`SFT-MATH-GRAPH-NETWORK-001`	512	9	depth-independent
5	`SFT-MATH-ALGEBRA-001`	512	9	depth-independent
6	`SFT-MATH-ORDER-LATTICE-001`	512	9	depth-independent
7	`SFT-MATH-GEOMETRY-TOPOLOGY-001`	1,024	10	depth-independent
8	`SFT-MATH-PROBABILITY-STATISTICS-001`	512	9	depth-independent
9	`SFT-MATH-OPTIMIZATION-001`	512	9	depth-independent
10	`SFT-MATH-DYNAMICAL-SYSTEMS-001`	1,024	10	depth-independent

Order	Claim	Candidate structures	Dimensions	Closure
11	`SFT-MATH-LOGIC-PROOF-001`	1,024	10	depth-independent
12	`SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`	1,024	10	depth-independent

The Mathematics total is **7,424** generated structures and twelve survivors. The full V3 census, including the ten Foundation claims, contains 22 admitted derivations and 9,874 generated candidate structures across the two complete branches.

The finite product count reports representation classes executed by the registered claim grammars. It is not a count of every mathematical object producible by the laws. Generality is carried by the induction certificate stated for each claim below.

6. Derivation 1: Exact arithmetic and generated number structure

Claim identity: `SFT-MATH-EXACT-ARITHMETIC-001`

6.1 Question and necessity

Arithmetic is required before later branches can count arrangements, compare structures or assign exact support weights, but importing a conventional number field would violate the clean-room boundary.

The exact theorem statement is:

> *Every admitted arithmetic relation on generated positive finite traces and exact parts has one parameter-free structural kernel: disjoint junction for addition, complete pair-cell refinement for product, finite repeated composition for powers, common-refinement pairing for exact quotient and comparison, and held orientation for unmatched remainder; no semantic zero, negative, irrational, floating or completed-infinite value enters the law.*

Admitted dependencies: `SFT-FOUNDATION-COUNT-001`, `SFT-FOUNDATION-PART-001`, `SFT-FOUNDATION-PART-EQUIVALENCE-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`.

6.2 Generated grammar and exact boundary

Count supplies complete positive traces; Part supplies held/whole coordinates; Part Equivalence supplies common refinement and bijection. Exhausting the eight representation choices leaves one kernel that preserves every source identity without introducing a scale or forbidden value.

Generation rule: Generate the complete product of trace coverage, junction, pair-cell product, exact quotient, comparison, remainder orientation, finite generality and extra-rule status.

Exact grammar boundary: All arithmetic kernels constructible from admitted positive finite traces, exact held/whole parts, common refinement, canonical form identity and no imported scalar field.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:e5594c9df6b591e6d7bac2b5afb86bcb32a41d450fb06727426740b982a57500`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-EXACT-ARITHMETIC-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`coverage`	`partial-trace`	A partial trace omits an admitted generated occurrence.	`complete-trace` - The complete trace retains every generated occurrence once.
`junction`	`overlap-merge`	Overlap repeats a shared occurrence and loses disjoint provenance.	`disjoint-junction` - Disjoint junction preserves both complete positive traces.
`product`	`repeated-unlabelled`	Unlabelled repetition erases the two source coordinates.	`pair-cell-refinement` - Each generated left label is paired once with each generated right label.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`quotient`	`rounded-measure`	A rounded or decimal value is measured rather than exact derivational evidence.	`common-refinement-pairing` - A complete common refinement and bijection witnesses the exact part relation.
`comparison`	`borrowed-scalar-order`	A borrowed scalar order imports the answer-producing number model.	`complete-trace-pairing` - Pairing identifies equality and a held unmatched trace identifies orientation.
`difference`	`signed-magnitude`	A signed magnitude installs forbidden negative quantity.	`held-oriented-remainder` - A distinct held label records which trace retains the unmatched positive remainder.
`generality`	`bounded-answer-table`	A finite answer table has no depth-independent arithmetic certificate.	`base-successor-induction` - The One base and generated successor preserve each operation at every finite depth.
`addition`	`has-extra-rule`	An extra scale, constant or field is not supplied by the dependencies.	`no-extra-rule` - The kernel uses only admitted traces, parts, refinement, pairing and held labels.

6.3 Unique survivor, minimality and laws

> *The exact arithmetic kernel is complete positive-trace coverage with disjoint junction, pair-cell product, common-refinement quotient, complete pairing comparison, held-oriented remainder, base/successor generality and no extra rule.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- disjoint junction is associative on mutually disjoint generated traces.
- pair-cell product distributes over disjoint junction by generated cell provenance.
- pair-cell product composition is associative after canonical path flattening.
- exact quotient exists only with a complete refinement-pairing witness.
- comparison returns same-form or a held positive remainder orientation.

6.4 Operational witnesses and unfavorable checks

- `junction-associativity` - (a joined b) joined c and a joined (b joined c) retain the same canonical trace. Result: PASS.
- `pair-cell-cardinality` - Two generated cells paired with three generated cells produce the complete six labelled pair cells. Result: PASS.
- `product-associativity` - Canonical flattened pair paths are independent of binary grouping. Result: PASS.
- `exact-pairing` - Equal complete traces admit one-to-one and onto pairing; unequal traces do not. Result: PASS.
- `held-remainder` - A larger trace reports a positive held remainder instead of a negative value. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial trace omits an admitted generated occurrence. Result: PASS; receipt `sha256:33e3d5d241540fd337807b34e03db365ce48e9e3d0ea4d102d153a58df8199c6`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.

- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not a value; negative magnitude is replaced by held orientation; irrational, imaginary and floating proof values are excluded; completed infinity and an ungenerated continuum are excluded Result: PASS; receipt `sha256:19138ef235a0e43f75285e80856edccff9d19b3aced53986c83f5de9fa2f4fca`.

6.5 Depth-independent closure

Base: The structural One supplies the first nonempty trace and exact self-whole.

Successor: Appending one fresh generated occurrence preserves complete trace identity; junction appends it once, pair-cell refinement appends one labelled cell per opposite trace element, and pairing either extends with one mate or records the unmatched held remainder.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

6.6 Meaning and scientific consequence

This result changes the role of arithmetic. A number is not admitted as an unexplained occupant of a pre-existing field. Exact positive magnitude is the identity of a complete generated trace. Addition is the preservation of two disjoint traces in their junction; product is the complete provenance-retaining pair-cell refinement. An exact quotient is not a division oracle but the existence of a complete pairing over a common refinement. Comparison never requires a negative result: equality is complete pairing and difference is a held orientation together with the unmatched positive trace.

6.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: positive integers, rational numbers, addition, multiplication, division, order, signed difference. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

6.8 Exact exclusions and limitations

- semantic numerical zero is not a value.
- negative magnitude is replaced by held orientation.
- irrational, imaginary and floating proof values are excluded.
- completed infinity and an ungenerated continuum are excluded.

This law closes exact generated finite arithmetic and positive rational part relations. It does not admit a completed infinite set, continuum, irrational field or negative proof magnitude. Conventional notation may describe the result only in correspondence.

6.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:db8d9a5838f2f84cd04cd3425e0faa5232ec8afb6461f1a7fe62ab9f6263246f`.
- Derivation seal: `sha256:1254abf40a91886525ad8a846896b75a1023fd785907f233bcf817db2c744dc4`.
- Independent implementation: `sha256:9d0af694427b46f2b09ed218d83d0d87f30b16601b03c3694ac71738254bf164`.
- Independent certificate: `sha256:73b24906c1005bd7f42f36000f396bb5ec88f633a4a69b6159d5cc349bb3be2a`.
- External validation: `sha256:9a05361b10ccce837c8a32def2b24dd0ec714cb2f45d397a9e6b72fedec306a7`.
- Engine receipt: `sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418`.

- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-EXACT-ARITHMETIC-001-28252bae62373d86.json`.

7. Derivation 2: Generated discrete objects, relations and induction

Claim identity: `SFT-MATH-DISCRETE-001`

7.1 Question and necessity

Combinatorics, graphs, algebra, logic and computation require exact collections and relations, but an imported set-theoretic universe would add axioms and completed infinities outside the SFT domain.

The exact theorem statement is:

> *Every admitted discrete object is a complete canonical generated finite collection or held selection; membership is retained occurrence, relations are held selections of complete pair-cell support, maps are total single-valued relations, sequences retain generated order, and general laws close by structural One/base-successor induction.*

Admitted dependencies: `SFT-FOUNDATION-COUNT-001`, `SFT-FOUNDATION-FORM-GRAMMAR-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-EXACT-ARITHMETIC-001`.

7.2 Generated grammar and exact boundary

Canonical form enforcement supplies identity; count supplies complete finite traces; exact arithmetic supplies lawful trace composition. Product enumeration of the eight representational questions forces the collection-selection-relation-map-induction kernel.

Generation rule: Generate the complete product of collection coverage, canonical identity, membership, relation support, map law, sequence order, induction closure and extra-constructor status.

Exact grammar boundary: All discrete structures generated from finite canonical One/Fold forms, exact positive traces, held selections, pair-cell relations and successor construction.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:0103778e8716a64ec9ed8d8b243af9af9bc7fed90360339b4be989a03d263302`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-DISCRETE-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`collection`	`partial-or-duplicated`	Omission or duplication changes the generated collection trace.	`complete-unique-trace` - Every generated member occurs once in the canonical collection trace.
`identity`	`presentation-alias`	Presentation names can merge distinct forms or split one form.	`canonical-form-identity` - Canonical construction traces supply exact member identity.
`membership`	`unrecorded-assertion`	An unrecorded assertion has no source-bound occurrence witness.	`retained-occurrence` - Membership retains the canonical member occurrence within the source trace.
`relation`	`free-pair-list`	A free pair list has no complete source-product boundary.	`held-pair-cell-selection` - A relation holds a generated selection from complete source pair-cell support.
`map`	`partial-or-multivalued`	A missing or repeated image does not define one image for every source member.	`total-single-valued` - Every source member has exactly one retained target mate.
`sequence`	`unordered-carrier`	An unordered carrier erases the generated predecessor relation.	`held-generation-order` - The canonical trace retains each predecessor-successor position.
`induction`	`sampled-depths`	Sampled depths do not establish the successor step.	`One-base-successor` - The One base and freshness-preserving successor cover every generated finite trace.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`addition`	`extra-constructor`	An extra constructor is not supplied by the Foundation grammar.	`no-extra-constructor` - All objects decompose into admitted forms, selections, pair cells and successors.

7.3 Unique survivor, minimality and laws

> *The discrete kernel is complete unique canonical collections with retained membership, pair-cell-selected relations, total single-valued maps, held sequence order, One/base-successor induction and no extra constructor.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- canonical membership is decidable for every generated finite collection.
- held selection never introduces an element outside its source collection.
- relation complement is a held complementary selection of the same complete pair support.
- map composition preserves total single-valuedness when interfaces agree.
- structural induction covers every generated finite collection trace.

7.4 Operational witnesses and unfavorable checks

- `unique-collection` - A generated collection retains each canonical member once. Result: PASS.
- `held-membership` - A held selection contains only members of its registered source. Result: PASS.
- `empty-One-selection` - A held-empty selection returns the empty One form, not numerical zero. Result: PASS.
- `relation-boundary` - Every retained relation pair belongs to the complete generated pair-cell support. Result: PASS.
- `total-map` - The example relation gives every source form exactly one image. Result: PASS.
- `fresh-successor` - A successor appends exactly one fresh canonical form. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or duplication changes the generated collection trace. Result: PASS; receipt `sha256:bfcb72c0f0a5cfdb140547bb18d237268d4ae9231dbf57f6be92112190ed979a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no completed infinite set; no ungenerated universal collection; no semantic numerical zero for the empty selection; no presentation alias may replace canonical form identity Result: PASS; receipt `sha256:fdcae55e3e8ce516575f15e1d490505002d0dffbdc3dc372c54ed70cd69b236e`.

7.5 Depth-independent closure

Base: The structural One is the first canonical generated member and the empty One marks a held-empty selection without numerical zero.

Successor: Given a complete unique canonical collection, append one fresh canonical form once; membership, relation pair support, map-image obligation and predecessor order extend by their local generated clauses.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

7.6 Meaning and scientific consequence

The theorem reconstructs the working core normally supplied by an ambient set universe. A collection has a finite generated boundary, canonical member identities and no unrestricted comprehension. Membership is a retained occurrence; a relation is a held selection from complete pair support; and a map is a relation that passes totality and single-valuedness. This makes every later discrete object auditable at its source.

7.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: finite set, subset, membership, relation, function, sequence, structural induction. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

7.8 Exact exclusions and limitations

- no completed infinite set.
- no ungenerated universal collection.
- no semantic numerical zero for the empty selection.
- no presentation alias may replace canonical form identity.

The theorem concerns generated finite objects. It does not install an ungenerated universe, completed infinite set or unrestricted comprehension principle.

7.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:41a82043fd37d0c054d490f9c426ad37c23d6bb8ea2e4f0e9a5af6fb8fe30285`.
- Derivation seal: `sha256:1d3b73c941e235c5ffe473ccafa9cd59b800ddd33d43e4feea7a6821ed8f45c9`.
- Independent implementation: `sha256:c808241fc28e1fa7bfbbea0e67fc594eab9b957575534f2f0b88af57def8dbae`.
- Independent certificate: `sha256:2f9da4efdc67b6f08074be8d3ddc47595444a9300fd00e507fb04c228686f44b`.
- External validation: `sha256:5dad446dc09d74caf9325f27b314d7f886275164df426cd561a687d9283cd109`.
- Engine receipt: `sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-DISCRETE-001-632de46c995329ed.json`.

8. Derivation 3: Exact finite combinatorial generation

Claim identity: `SFT-MATH-COMBINATORICS-001`

8.1 Question and necessity

Counting arrangements by quoting familiar formulas would import their answers. The generated family itself must be primary so every count, symmetry reduction and recurrence remains inspectable.

The exact theorem statement is:

> *Every admitted finite combinatorial family is generated by an explicit held-choice recurrence over a complete canonical carrier, preserves construction order and provenance, removes duplicates only by canonical form identity, and counts the resulting complete trace; selections include the structural empty One form rather than numerical zero.*

Admitted dependencies: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`.

8.2 Generated grammar and exact boundary

Discrete canonical collections provide the carrier and held selection; Fold words provide exact choice traces; arithmetic counts the complete result. Exhausting the eight assembly choices forces explicit held-choice recurrence and provenance-preserving class accounting.

Generation rule: Generate the complete product of carrier coverage, choice rule, provenance, duplication policy, symmetry identity, class accounting, recurrence closure and extra-formula status.

Exact grammar boundary: All arrangement, selection and finite classification families produced from a registered canonical finite carrier by held choice, remainder and successor recurrence.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:b5a4fe7ce476fb3f349f0ac30a1ccfbd35c2637505be5ef9c4022d90e65a403b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-COMBINATORICS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`partial-carrier`	A partial carrier silently omits possible generated choices.	`complete-canonical-carrier` - Every distinct generated source member is present once.
`choice`	`sampled-choices`	Sampling does not exhaust the registered choice positions.	`held-choice-recurrence` - Each available member is held once at the next position and recursion continues on the exact remainder.
`provenance`	`erased-origin`	Erasing origin prevents reconstruction and duplicate diagnosis.	`complete-choice-trace` - Every result retains the ordered held-choice construction trace.
`duplicates`	`presentation-deduplication`	Presentation equality can merge structurally distinct results.	`canonical-identity-deduplication` - Only identical canonical construction traces are identified.
`symmetry`	`assumed-symmetry`	An assumed quotient may remove distinguishable held labels.	`generated-bijection-witness` - A declared label-preserving bijection is required before forms share a symmetry class.
`accounting`	`signed-inclusion-subtraction`	Signed subtraction imports negative quantities.	`disjoint-provenance-classes` - Each generated form is assigned once to a held disjoint class before counts are joined.
`generality`	`finite-size-table`	A table of small sizes does not force the next recurrence.	`carrier-successor-recurrence` - Adding one fresh carrier member partitions new forms by whether and where it is held.
`addition`	`imported-formula`	A stored factorial, binomial or inclusion formula selects the conventional answer.	`no-imported-formula` - Counts are identities of the complete generated output traces.

8.3 Unique survivor, minimality and laws

> *The combinatorial kernel is complete canonical carrier generation by held-choice recurrence with complete choice provenance, canonical deduplication, witnessed symmetry, disjoint class accounting, successor generality and no imported formula.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- every arrangement holds each carrier member exactly once.
- every selection is produced once by the fresh-member absent/held recurrence.
- canonical trace equality is the only duplicate-elimination rule.
- disjoint class ledgers conserve the complete generated family.
- enumerated count is the identity of the complete output trace.

8.4 Operational witnesses and unfavorable checks

- `arrangement-completeness` - Three distinct held labels generate six unique ordered arrangements. Result: PASS.
- `arrangement-coverage` - Every arrangement holds every carrier label exactly once. Result: PASS.
- `selection-recurrence` - Three held labels generate the structural empty One and every absent/held combination once. Result: PASS.
- `disjoint-accounting` - Parity-named witness classes partition the carrier without overlap or omission. Result: PASS.
- `fresh-extension` - Fresh-member recurrence creates a disjoint absent/held copy of every prior selection. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial carrier silently omits possible generated choices. Result: PASS; receipt `sha256:c1dbfffb55e34b1990cfc2d3bd521fd163d147f13c5598b09021959fcee3cfe3`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no random sampling as completeness evidence; no factorial or binomial answer table; no negative inclusion-exclusion proof value; no completed infinite combinatorial family Result: PASS; receipt `sha256:5e256307e37a27f3cb51ece4a149427a8131c602ec3eb31051eb5b10fdb3dced`.

8.5 Depth-independent closure

Base: A one-member carrier has one complete arrangement and two structural selection forms: empty One and held member.

Successor: For a fresh carrier member, arrangements partition by its held position and recurse on the prior remainder; selections partition into prior forms and the same forms with the fresh member held. These disjoint generated classes exhaust the successor carrier without an imported formula.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

8.6 Meaning and scientific consequence

Combinatorial values arise after, not before, generation. The arrangement and selection families are constructed by exhaustive held choices with each remainder retained. Their counts are identities of the resulting complete traces. Symmetry reduction requires an explicit label-preserving bijection, and overlap is handled by disjoint provenance classes instead of negative inclusion-exclusion proof quantities.

8.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: permutation, combination, recurrence, symmetry class, pigeonhole counting, inclusion-exclusion. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

8.8 Exact exclusions and limitations

- no random sampling as completeness evidence.
- no factorial or binomial answer table.
- no negative inclusion-exclusion proof value.
- no completed infinite combinatorial family.

The law closes the generative foundation of finite combinatorics. Named advanced enumerations require their own derived generators; none may be admitted by quoting a conventional closed-form answer.

8.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:7d7554f879a30b739c05c4f9aeba7c5812c4ceddea2a1d032a71ecb8fa10ba1a`.
- Derivation seal: `sha256:3177572ea372062f65a7969fdf4e498c86c9af558e37c048165e1d950cf1d942`.
- Independent implementation: `sha256:00ac1dfb12aa98fba2ec8b0ab7eae0b940cee7ca8c55b75fa07678864e9b4086`.
- Independent certificate: `sha256:ea22b95c87c9f336726c9df87d1e88fb38b1ad27519789c334a5052f5f864ae6`.
- External validation: `sha256:dbfc2711bbabde9be3bf4a4cc3c5de3cd62440c9fbeea20b2fceed6dcfb1275b`.
- Engine receipt: `sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-COMBINATORICS-001-52e1db94bb786527.json`.

9. Derivation 4: Exact finite graph and network structure

Claim identity: `SFT-MATH-GRAPH-NETWORK-001`

9.1 Question and necessity

Graphs are the minimal mathematics of explicit relations and paths. Their laws must follow from generated carriers and pair support rather than imported matrices, weights or probabilistic ensembles.

The exact theorem statement is:

> *Every admitted finite graph is a complete canonical node collection plus a held selection of generated ordered node-pair relations; paths retain every adjacent transition, reachability is a complete finite path witness, cycles retain return, trees are connected cycle-free forms, cuts are exact crossing-edge selections, and internal network conservation is complete ingress/egress pairing.*

Admitted dependencies: `SFT-FOUNDATION-FORM-GRAMMAR-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-COMBINATORICS-001`.

9.2 Generated grammar and exact boundary

Discrete mathematics supplies canonical carriers and relations; combinatorics supplies complete path-family generation. Exhausting nine structural choices leaves the relation/path/return/pairing graph kernel.

Generation rule: Generate the complete product of node coverage, edge provenance, orientation, path trace, connectivity, cycle return, cut support, internal balance and extra-network-rule status.

Exact grammar boundary: All finite graphs and networks generated from canonical finite form carriers, held pair-cell relations, complete path traces, return relations and exact positive flow-token pairings.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:447b14b12f08e4bcf5c21bd3b34a7e06cbccd3ad90d3669153c9d05306221bfd`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-GRAPH-NETWORK-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`nodes`	`partial-or-aliased-nodes`	Missing or aliased nodes change the graph carrier.	`complete-canonical-nodes` - Every generated node form occurs once with canonical identity.
`edges`	`free-endpoint-list`	A free list can contain ungenerated endpoints or duplicate relations.	`held-pair-cell-relation` - Edges are a held duplicate-free selection of complete node pair support.
`orientation`	`signed-edge-weight`	A signed weight imports negative magnitude and does not identify endpoints.	`held-endpoint-order` - The ordered endpoint labels retain orientation; complementary order represents reversal.
`path`	`endpoint-only`	Endpoints alone omit the intervening adjacency requirements.	`complete-adjacent-trace` - Every node and edge transition in the path is retained in order.
`connectivity`	`assumed-connected`	An assertion without a path omits the connecting structure.	`generated-path-witness` - Reachability requires a complete generated adjacent path.
`cycle`	`repeated-label-only`	A repeated label without a transition trace does not prove return.	`explicit-return-path` - A cycle is a nontrivial complete path whose terminal relation returns to a held prior node.
`cut`	`borrowed-scalar-threshold`	A scalar threshold imports an external coordinate.	`held-crossing-selection` - A held node selection forces exactly the edges with one held and one complementary endpoint.
`balance`	`unaccounted-net-value`	A net signed value hides unmatched tokens.	`ingress-egress-pairing` - Every incoming positive token has one outgoing mate and conversely.
`addition`	`extra-network-rule`	An extra weight, metric or random law is not supplied by the graph dependencies.	`no-extra-network-rule` - All graph properties are witnessed by carriers, relations, paths, returns, selections and pairings.

9.3 Unique survivor, minimality and laws

> *The graph/network kernel is complete canonical nodes with held ordered pair-cell edges, complete adjacent paths, witnessed reachability and return, held crossing cuts, ingress/egress pairing and no extra rule.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- path concatenation is lawful exactly when the interface node agrees.
- reachability is reflexive by the retained identity path and transitive by path composition.
- a tree is a connected generated graph with no nontrivial return path.
- a cut contains every and only relation crossing a held/complementary node partition.
- network balance preserves token identity rather than cancelling signed magnitudes.

9.4 Operational witnesses and unfavorable checks

- `valid-path` - The chain retains both required adjacent edges. Result: PASS.
- `missing-edge-control` - An endpoint sequence with a missing adjacency is not a path. Result: PASS.
- `reachability` - Generated path search reaches c from a and does not reach a from c in the directed chain. Result: PASS.

- `cycle-return` - The explicit c-to-a return distinguishes the cycle from the chain. Result: PASS.
- `cut-selection` - Holding the first chain node selects exactly its crossing edge. Result: PASS.
- `paired-balance` - Equal unique ingress and egress token traces witness internal balance. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Missing or aliased nodes change the graph carrier. Result: PASS; receipt `sha256:2c881c5d8c3f3ab8e894f11a4fce432fbbfa4e49226ddb271ea8023ed8e3228b`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ungenerated vertex or edge; no negative edge or flow quantity; no stochastic graph law; no infinite graph as a completed object Result: PASS; receipt `sha256:3f199f993e865f79dd3ff17ba48418bbfca2470b4a79c8a8697148409b3ba7be`.

9.5 Depth-independent closure

Base: One canonical node supplies the identity path and contains no nontrivial edge cycle.

Successor: Appending one fresh node generates exactly its ordered pair cells with prior nodes; any held edge selection extends paths locally, and connectivity, cycle, cut and balance witnesses update only through those new cells.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

9.6 Meaning and scientific consequence

Graph theory becomes the exact mathematics of generated relations. Vertices are canonical forms, edges are held ordered pair cells, and paths preserve every adjacency. Reachability, cycles and trees are not labels but path predicates. A cut is forced by a held/complementary node selection. Network conservation retains the identities of paired ingress and egress tokens instead of cancelling signed scalar totals.

9.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: directed graph, path, reachability, cycle, tree, cut, network flow. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

9.8 Exact exclusions and limitations

- no ungenerated vertex or edge.
- no negative edge or flow quantity.
- no stochastic graph law.
- no infinite graph as a completed object.

This law closes exact generated finite graph structure. Weighted geometry, random graphs, infinite graph limits and algorithmic graph complexity require later derived laws.

9.9 Evidence identities

- Registered status: `independently_replicated`.

- Source manifest: `sha256:e9e22d5262250c7cb551e278121b29431ef59d58cb53b28c9188f2d912979d8c`.
- Derivation seal: `sha256:f1f74c52072189f08134293f1f55aaff63aeb73fc045d79ac09a9372554c7ad9`.
- Independent implementation: `sha256:82cc03abc10d06d9d288a85c6171bc60db065b82a074e9fc39745f5b1e59ebe4`.
- Independent certificate: `sha256:531d0c9a5ec7a97b8e97c4a1664d67af65d7fab17127bcf042e76a86386a9210`.
- External validation: `sha256:9f9e2483be55ff773cf570f6fc3aee074f68f44fde118492f152023767fef5c8`.
- Engine receipt: `sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-GRAPH-NETWORK-001-b4072f93147c8cd9.json`.

10. Derivation 5: Generated finite algebraic structures

Claim identity: `SFT-MATH-ALGEBRA-001`

10.1 Question and necessity

Algebra is forced when generated operations are composed and compared. The exhaustive table boundary prevents familiar algebraic classifications from supplying unproved laws.

The exact theorem statement is:

> *An admitted finite algebraic structure is a complete canonical carrier with a complete, single-valued, closed pair-cell operation relation. Identity, associative composition, reversible return mates, commutation, substructure and structure-preserving maps are admitted exactly when their exhaustive carrier witnesses pass; none is installed as an ungenerated axiom.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`.

10.2 Generated grammar and exact boundary

Discrete pair support supplies all operation inputs; exact canonical equality tests results; form enforcement supplies carrier identity. The nine-way census leaves the witness-governed operation kernel.

Generation rule: Generate the complete product of carrier, operation coverage, closure, identity, association, return, property-admission, structure-map and extra-law status.

Exact grammar boundary: All finite operation structures generated from canonical carriers, complete pair-cell relations, canonical path composition, held reversal and exhaustive property witnesses.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:ccae1cb3b71ff2b6cffd4acead871b01722969c069b797ad61e9cb2050f09701`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ALGEBRA-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`partial-or-aliased-carrier`	Omission or aliasing changes which operation inputs exist.	`complete-canonical-carrier` - Every generated carrier form occurs once with canonical identity.
`operation`	`partial-operation-sample`	A partial sample does not assign every generated input pair.	`complete-single-valued-table` - Every ordered carrier pair has exactly one retained result.
`closure`	`external-result`	An external result silently enlarges the registered carrier.	`carrier-closed-result` - Every result is a canonical member of the same carrier.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`identity`	`named-identity`	A name alone does not prove left and right preservation.	`exhaustive-identity-witness` - One carrier form preserves every member on both sides.
`association`	`assumed-parenthesis-law`	An assumed equation imports an algebraic axiom.	`complete-triple-witness` - Every carrier triple has equal canonical flattened composites.
`return`	`signed-inverse-value`	A signed magnitude imports negative quantity.	`unique-held-return-mate` - A retained complementary carrier mate returns on both sides to identity.
`properties`	`property-by-name`	A familiar structure name cannot supply commutation or return.	`property-by-exhaustive-witness` - Each property is registered only after every required carrier tuple passes.
`maps`	`arbitrary-carrier-map`	An arbitrary map can erase the operation relation.	`operation-preserving-map` - The complete image relation commutes with every source operation cell.
`addition`	`extra-algebraic-axiom`	An extra axiom is not forced by the registered operation traces.	`no-extra-algebraic-axiom` - All admitted properties have explicit generated witnesses.

10.3 Unique survivor, minimality and laws

> *The algebraic kernel is a complete canonical carrier and complete closed single-valued operation table, with identity, association, returns, special properties and maps admitted only by exhaustive witnesses.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- closure is exact membership of every operation result in the registered carrier.
- identity is a two-sided exhaustive preservation witness.
- association is equality of both complete composite traces for every carrier triple.
- return is a unique held mate relation and never a negative proof value.
- homomorphism is complete operation preservation, not resemblance of presentations.

10.4 Operational witnesses and unfavorable checks

- `closed-table` - Every complete input pair in the witness table has one carrier result. Result: PASS.
- `identity` - The structural One form preserves both witness carrier forms. Result: PASS.
- `association` - Both parenthesizations agree for every witness triple. Result: PASS.
- `return-mates` - Every witness form has one held two-sided return mate. Result: PASS.
- `commutation-is-conditional` - Commutation passes for the symmetric witness and fails for a lawful noncommuting table. Result: PASS.
- `structure-map` - The canonical self-map preserves every operation cell. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or aliasing changes which operation inputs exist. Result: PASS; receipt `sha256:b0cb6ed9032c814991aae189ba084f58a0344948ecdeba10de30448360699e08`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.

- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported group, ring or field axioms; no negative or irrational carrier requirement; no property inferred from a conventional name; no completed infinite carrier Result: PASS; receipt `sha256:e7630fd960384d7d35b15f2e1c900d720e648b0dd23ec4cea7ac6ce9dadc209c`.

10.5 Depth-independent closure

Base: The One carrier has one pair cell, its result remains One, and identity and association coincide.

Successor: Adding one fresh carrier form generates exactly its left, right and self pair cells. A proposed extension is admitted only after results remain in the extended carrier and every newly formed identity, triple, return and map obligation is checked.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

10.6 Meaning and scientific consequence

The algebra theorem deliberately separates an operation kernel from named algebraic species. A complete closed operation table is forced first. Identity, association, return, commutation and homomorphism status are then admitted only if their complete carrier obligations pass. Thus the words group, monoid or commutative structure cannot import properties; they are correspondence classifications of an already sealed witness.

10.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: magma, monoid, group, commutativity, subalgebra, homomorphism. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

10.8 Exact exclusions and limitations

- no imported group, ring or field axioms.
- no negative or irrational carrier requirement.
- no property inferred from a conventional name.
- no completed infinite carrier.

This closes the finite algebraic kernel and its property tests. Particular richer algebraic species require their own generated carriers and independently witnessed additional operations.

10.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:c45fbdd44b40d4d5d007c619fbd6dd03554a43e980396d1d3158b5f9791b34a3`.
- Derivation seal: `sha256:ae92defb9a52942af89734332b791cb1921ae655499b56e808a6250f539d6396`.
- Independent implementation: `sha256:280231864879a04285ec75e17ecf961cb20c43bcf39c228fabf05b3ccd58e033`.
- Independent certificate: `sha256:b41bcba0ab04cfad3eb95cd55ae4286b9ea1c52748ae444b33e2330ec7b52bf4`.
- External validation: `sha256:5443f0a807dfeff488bfd29b7059fbbd086799f212e974d12867670b9f644654`.
- Engine receipt: `sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ALGEBRA-001-1f08bfaf9f602a38.json`.

11. Derivation 6: Exact finite order and conditional lattice structure

Claim identity: `SFT-MATH-ORDER-LATTICE-001`

11.1 Question and necessity

Order is needed for comparison, optimization, proof strength and topology. Retaining incomparability prevents a conventional scalar line from being imported where the Fold structure does not force one.

The exact theorem statement is:

> *An admitted finite order is a held carrier relation whose reflexive, antisymmetric and transitive cells are exhaustively witnessed. Totality is admitted only when every generated pair is comparable. Lower and upper bounds are complete relation selections; meet and join exist only as unique greatest-lower and least-upper witnesses, and monotone maps preserve every retained comparison.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-ALGEBRA-001`.

11.2 Generated grammar and exact boundary

Discrete relations supply the comparison cells; algebra supplies lawful composition; exact identity supplies antisymmetry. Exhausting nine choices forces witnessed partial order and conditional lattice operations.

Generation rule: Generate the complete product of carrier identity, relation provenance, partial-order laws, comparability, bound generation, meet uniqueness, join uniqueness, monotonicity and extra-order status.

Exact grammar boundary: All finite order structures generated from canonical carriers, held pair-cell relations, exhaustive relation closure, complete bound selections and unique extremal witnesses.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:06bc1ab2205eecb2fd980877791e2617c87d47e067be6162d904c903370cfeec`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ORDER-LATTICE-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`presentation-carrier`	Presentation aliases do not fix exact ordered identity.	`canonical-carrier` - Every ordered form has canonical generated identity.
`relation`	`borrowed-scalar-comparison`	A scalar comparison imports an answer-producing order.	`held-pair-relation` - Each comparison is a retained generated carrier pair.
`order`	`named-order`	A name does not establish reflexivity, antisymmetry and transitivity.	`exhaustive-partial-order-witness` - Every required relation cell and composite is checked.
`comparability`	`assumed-totality`	Partial orders can contain incomparable forms.	`pairwise-totality-witness` - Totality is recorded only when every generated pair is comparable.
`bounds`	`selected-bound`	Selecting a familiar candidate can omit another held condition.	`complete-bound-selection` - Every carrier form is tested against every held target.
`meet`	`assumed-meet`	A lower bound need not be greatest or unique.	`unique-greatest-lower-witness` - Exactly one lower bound is above every other lower bound.
`join`	`assumed-join`	An upper bound need not be least or unique.	`unique-least-upper-witness` - Exactly one upper bound is below every other upper bound.
`maps`	`arbitrary-map`	An arbitrary map can reverse a retained comparison.	`comparison-preserving-map` - Every source relation cell maps to a target relation cell.
`addition`	`extra-order-axiom`	An extra totality or completeness axiom is not structurally forced.	`no-extra-order-axiom` - Only exhaustively witnessed order properties are admitted.

11.3 Unique survivor, minimality and laws

> *The order/lattice kernel is a canonical carrier with an exhaustively witnessed partial-order relation, conditional totality, complete bound selections, uniquely witnessed meet/join and monotone maps.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- partial order requires reflexive, antisymmetric and transitive relation closure.
- incomparability is retained and never forced into a borrowed total scale.
- meet and join are partial operations unless unique extremal witnesses exist.
- a finite lattice is exactly an admitted partial order with meet and join for every generated pair.
- monotonicity is preservation of every registered relation cell.

11.4 Operational witnesses and unfavorable checks

- `partial-order` - The diamond relation passes all three partial-order obligations. Result: PASS.
- `incomparability-retained` - The diamond retains left and right as incomparable. Result: PASS.
- `conditional-totality` - The explicit chain is total while the diamond is not. Result: PASS.
- `meet` - The two incomparable diamond arms have structural One as their unique meet. Result: PASS.
- `join` - The two incomparable diamond arms have whole as their unique join. Result: PASS.
- `monotone-identity` - The canonical identity map preserves every diamond comparison. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Presentation aliases do not fix exact ordered identity. Result: PASS; receipt `sha256:8b99f164faa8fd4a22432325f9d4c0afd5c8882ed01e4d35781cbb3544f51254`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported real-number line; no assumed total order; no lattice completeness without generated meets and joins; no completed infinite chain Result: PASS; receipt `sha256:484dd6f1f271f8054f6f3e5e74696abdbe274c36cb08b223628218b599f5da79`.

11.5 Depth-independent closure

Base: The One carrier has its identity relation and One is both unique lower and upper bound of itself.

Successor: Adding one fresh canonical form generates all comparisons to prior forms. Order closure, comparability, bounds and extremal uniqueness are rechecked only for pairs and composites touching the fresh form.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

11.6 Meaning and scientific consequence

Order is relational, not borrowed from a numerical line. Reflexivity, antisymmetry and transitivity are exhausted over the carrier. Incomparability is positive retained information and is not forced into a total ranking. Bounds are complete selections. A meet or join exists only when the respective greatest-lower or least-upper witness is unique, so lattice structure remains conditional on generated evidence.

11.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: partial order, total order, poset, meet, join, lattice, monotone map. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

11.8 Exact exclusions and limitations

- no imported real-number line.
- no assumed total order.
- no lattice completeness without generated meets and joins.
- no completed infinite chain.

This theorem closes finite generated orders. It does not assert totality, lattice status or completeness for a carrier unless the corresponding finite witness is present.

11.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:fb4627050dca9b64d628dfc773d8de1d30656f40689d1cb8a0ae8dd3105940bd`.
- Derivation seal: `sha256:d4421e8aaf34c239bda0fcacfe2a6aa770fac55ec3afe50c0ba5085d83946527`.
- Independent implementation: `sha256:e0b750be248ae4a9da52c952d8988d01fa236ccdd6a994063a12f297d2088a8f`.
- Independent certificate: `sha256:f1b8cbb06e153828d4410413baf3b6af6f075df5a5f3f297a1f6dee4d1c51bce`.
- External validation: `sha256:f2b2f4bbab7729985711dbdff1cbc25cd7d16f0423a9752f8abdba8caebdd136`.
- Engine receipt: `sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ORDER-LATTICE-001-d0e20f40782f51cc.json`.

12. Derivation 7: Exact finite geometry and topology for computation

Claim identity: `SFT-MATH-GEOMETRY-TOPOLOGY-001`

12.1 Question and necessity

Algorithms need incidence, locality, paths, neighborhoods and continuity, but do not require importing a real-coordinate continuum. The finite structural form is the exact computationally required kernel.

The exact theorem statement is:

> *Computational geometry is forced as canonical finite cells plus an acyclic held boundary-incidence relation; dimension is retained boundary depth, adjacency is shared incidence, and distance is a shortest generated path with self-distance represented by empty One rather than numerical zero. Computational topology is a complete generated open-family closed under generated joins and pairwise intersections; continuity is exact inverse-image preservation.*

Admitted dependencies: `SFT-FOUNDATION-PART-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`.

12.2 Generated grammar and exact boundary

Graphs supply paths and incidence; orders supply joins and intersections; exact parts supply held/whole selection. The ten-dimensional census forces the finite incidence-and-open-family kernel.

Generation rule: Generate the complete product of cell carrier, incidence, dimension, adjacency, path distance, open-family closure, connectedness, continuity, deformation and extra-geometric status.

Exact grammar boundary: All finite incidence complexes, path geometries and finite open families generated from canonical forms, held boundary relations, exact path traces, selections, joins and intersections.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:be96c202d0f654cb38088fbd1c4afc210f949ce35b52525cdc7d3e45ad189ed0`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-GEOMETRY-TOPOLOGY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`cells`	`coordinate-samples`	Samples from a borrowed coordinate field omit structural provenance.	`canonical-generated-cells` - Every cell is an exact canonical generated form.
`incidence`	`assumed-containment`	Assumed containment has no exact face-to-cell witness.	`held-acyclic-boundary` - Every face relation is retained and boundary descent forbids return.
`dimension`	`borrowed-coordinate-count`	A coordinate count imports an ambient space.	`boundary-chain-depth` - Dimension is the longest retained face-descent trace.
`adjacency`	`metric-threshold`	A threshold imports a free scale.	`shared-incidence-witness` - Cells are adjacent exactly when they share a generated face.
`distance`	`floating-metric`	A floating metric imports precision and possibly irrational values.	`shortest-generated-path` - Distance is the complete least-length path trace; identity is empty One.
`opens`	`named-neighborhoods`	Named neighborhoods need not satisfy closure.	`generated-open-family-closure` - Empty One, whole, generated joins and pairwise intersections remain in the family.
`connectedness`	`visual-continuity`	A presentation cannot establish structural connection.	`no-disjoint-open-separation` - No two nonempty disjoint admitted opens jointly exhaust the carrier.
`continuity`	`epsilon-scale`	An imported scale is neither forced nor needed in the finite grammar.	`inverse-open-preservation` - Every target open has an admitted source inverse image.
`deformation`	`untracked-shape-change`	An untracked change can erase incidence.	`reversible-incidence-trace` - Each elementary change retains a lawful reverse and preserves registered incidence invariants.
`addition`	`continuum-or-extra-metric`	A continuum or metric field is outside the generated computational need.	`no-ambient-continuum` - The law uses only finite generated cells, paths and open families.

12.3 Unique survivor, minimality and laws

> *The geometry/topology kernel is finite canonical cells with held acyclic incidence, boundary-depth dimension, shared-face adjacency, exact shortest paths, closed finite open families, inverse-image continuity and reversible incidence-preserving deformation.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- cell dimension is determined by complete acyclic boundary descent.

- path distance retains a shortest generated transition trace and uses empty One at identity.
- finite topological closure is exhaustively decidable from the registered open family.
- continuity is exact inverse-image preservation of every admitted target open.
- geometric equivalence requires a reversible invariant-preserving transformation trace.

12.4 Operational witnesses and unfavorable checks

- `incidence` - Every witness face and cell belongs to the finite carrier. Result: PASS.
- `dimension-depth` - The witness triangle has one more boundary layer than its edge. Result: PASS.
- `adjacency` - The two witness edges are adjacent exactly through their shared q face. Result: PASS.
- `path-distance` - Shortest path retains a-to-b-to-c and identity returns empty One. Result: PASS.
- `finite-topology` - The complete two-point selection family is closed. Result: PASS.
- `continuity` - The identity map preserves every witness open by inverse image. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Samples from a borrowed coordinate field omit structural provenance. Result: PASS; receipt `sha256:602d7acdc1da5b0de53c819a78f381ccac9858cd0bba3688edd642448a2ff375`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ungenerated continuum; no irrational or floating proof coordinate; no free metric threshold; no completed infinite topological family Result: PASS; receipt `sha256:2133cc61a907c35bdc264f3d462617d553589263450769cb5527f812c4268469`.

12.5 Depth-independent closure

Base: One point is a canonical cell; its open family consists of empty One and the whole point carrier.

Successor: Adding one fresh cell generates its possible boundary incidences, shared-face adjacencies and path edges; adding one fresh point generates exactly the open selections needed for closure checks. Every new boundary, path, intersection, join and inverse image is then exhaustively tested.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

12.6 Meaning and scientific consequence

The theorem derives exactly the geometry and topology required by finite computation without an ambient continuum. Geometry is incidence, boundary depth, shared-face adjacency and shortest retained path. Topology is a generated family of opens closed under the operations available at the finite boundary. Continuity is inverse-open preservation, while deformation requires a reversible incidence-preserving trace. Smooth or real-coordinate descriptions may later correspond to approximations but do not select this kernel.

12.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: incidence complex, dimension, adjacency, metric, topology, continuity, homotopy. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

12.8 Exact exclusions and limitations

- no ungenerated continuum.
- no irrational or floating proof coordinate.
- no free metric threshold.
- no completed infinite topological family.

This closes geometry and topology where finite computation requires them. Smooth, differential, measure and continuum correspondences are not premises and require separate generated approximation laws if used.

12.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:36d95c3b1ee446c17b5b4e4f609e10ce1f679f164c720eebd8e482ab68873eaa`.
- Derivation seal: `sha256:7afff182127d9b93f9705212b997801eafd5a3629fd2a8a56753ce674832d0ad`.
- Independent implementation: `sha256:27f1aa8512b811cfef6ccf01fb51ebe844c79f99414e59d4783bc293980bb132`.
- Independent certificate: `sha256:8b8595f6ef14fa36929ff546752015da701e4a49af5213d54f0f6a1d4559ce68`.
- External validation: `sha256:35658ed3991ba98df705a53a75bacc5c7e813bde5f0505e5d20b549ed8cf3dd5`.
- Engine receipt: `sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-GEOMETRY-TOPOLOGY-001-a04f7de0ede6e234.json`.

13. Derivation 8: Exact finite probability and statistics from Fold observation

Claim identity: `SFT-MATH-PROBABILITY-STATISTICS-001`

13.1 Question and necessity

Prediction and experiment require uncertainty and statistics, but a superdeterministic Fold model cannot import stochastic dynamics. Exact observation classes supply the required epistemic structure.

The exact theorem statement is:

> *Probability is an exact held-support/whole-support part relation over a complete deterministic generated microstate support; uncertainty records distinctions closed by an observation class and is not a causal random premise. Conditional weight is exact common-refinement restriction, independence is complete pair-cell factorization, and statistics are provenance-retaining finite trace summaries. The empty event is empty One, never numerical zero.*

Admitted dependencies: `SFT-FOUNDATION-PART-EQUIVALENCE-001`, `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`.

13.2 Generated grammar and exact boundary

Parts supply exact held/whole quantities; combinatorics supplies complete support; the measurement boundary isolates data. Nine structural choices force the observation-support account without a prior.

Generation rule: Generate the complete product of support coverage, event formation, weight, uncertainty status, conditioning, independence, statistics, empirical isolation and extra-random-law status.

Exact grammar boundary: All exact finite probability spaces and statistical traces generated from complete microstate support, held selections, observation classes, common refinement, exact parts and sealed empirical measurements.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:0702345917ea32913e5c17e13d86f16d7a885f34e09f8941c7e90be0963f286b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-PROBABILITY-STATISTICS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`sampled-or-partial-support`	A partial support silently changes every derived weight.	`complete-generated-support` - Every registered deterministic microstate occurs once.
`event`	`named-outcome`	A name without retained membership does not identify support.	`held-support-selection` - An event is a duplicate-free held selection of exact support.
`weight`	`floating-propensity`	A floating propensity imports precision and a stochastic cause.	`exact-held-whole-part` - Weight is the exact held count over the complete support count.
`uncertainty`	`ontic-random-cause`	A causal random source is neither generated nor required.	`closed-observation-distinction` - Uncertainty records which deterministic microstate distinctions observation does not retain.
`conditioning`	`renormalized-guess`	A guessed renormalization can omit common support.	`exact-common-refinement` - Restrict to the complete held condition and count its exact event intersection.
`independence`	`uncorrelated-assertion`	An assertion does not prove factorized joint support.	`pair-cell-factorization` - Complete joint support is the exact product and event weights multiply by bijection.
`statistics`	`opaque-estimator`	An opaque scalar erases source rows and tie structure.	`provenance-retaining-summary` - Every summary is recomputable from the complete finite trace and retains ties.
`empirical`	`target-informed-fit`	Using target rows during derivation allows them to select the law.	`post-seal-blind-check` - Natural measurements enter only after formal sealing under registered isolation.
`addition`	`extra-random-parameter`	A seed distribution or prior is a free parameter.	`no-extra-random-parameter` - All weights follow from support and observation without stochastic dynamics.

13.3 Unique survivor, minimality and laws

> *The probability/statistics kernel is complete deterministic support, held events, exact held/whole weights, observation-class uncertainty, common-refinement conditioning, pair-cell independence, transparent finite summaries and post-seal empirical checking without random parameters.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- event weight is an exact positive part relation; empty event is empty One.
- conditional weight is exact restriction to a nonempty held condition.
- independence is a bijective complete joint-support factorization.
- uncertainty belongs to an observation partition and does not alter deterministic microstate evolution.
- statistical summaries retain exact provenance and all tied alternatives.

13.4 Operational witnesses and unfavorable checks

- `exact-weight` - Two held states in four complete states have the exact one-over-two part. Result: PASS.
- `empty-event` - The empty event is structural empty One rather than numerical zero. Result: PASS.
- `conditional` - Conditioning on the two a-prefix states retains one selected state as one-over-two. Result: PASS.

- `independence` - The complete two-by-two pair support factorizes exactly. Result: PASS.
- `ties-retained` - An exact modal summary preserves both equally supported alternatives. Result: PASS.
- `observation-classes` - A prefix observation retains two exact two-state classes. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial support silently changes every derived weight. Result: PASS; receipt `sha256:646eb932b825e9c344756ec999063ca07df35072a81f697e0b591039c293685a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ontic stochastic premise; no floating probability in proof evidence; no Bayesian or distributional prior parameter; no target data may select a formal law Result: PASS; receipt `sha256:6f88285aa8b2bb6873871ac8acd897d8ea1b6e120bfad06b20ca3c3f9ad50133`.

13.5 Depth-independent closure

Base: A One-state support has whole certainty; its held-empty event is represented by empty One.

Successor: Adding one fresh microstate extends whole support once and either extends or leaves each held event. Exact parts, observation classes, intersections and summaries update from that retained membership without a prior.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

13.6 Meaning and scientific consequence

Probability is compatible with superdeterminism because it is not installed as a random cause. The complete microstate support is deterministic; an observation closes some distinctions and retains others. Exact event weight is the held-support/whole-support part relation. Conditioning is exact restriction to a held common refinement, and independence is a bijective product-support fact. Statistics remain recomputable from all source rows and preserve ties. Natural data may test a sealed consequence but cannot choose it.

13.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: probability space, conditional probability, independence, random variable, statistic, sampling. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

13.8 Exact exclusions and limitations

- no ontic stochastic premise.
- no floating probability in proof evidence.
- no Bayesian or distributional prior parameter.
- no target data may select a formal law.

This closes exact finite probability and descriptive statistics. Inferential claims about natural data remain empirical claims and require preregistered blind validation through the empirical engine.

13.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:45f28c927c6994a7817ce2fd870441d41ee99e030be73063970674b092e6bab4`.
- Derivation seal: `sha256:9ad2c0aa860b03d0fe11c390e7e1fada396f48565fd89d055f06fec28b32ed67`.
- Independent implementation: `sha256:cdb15cdfaa90a02a1222dcbb0e8000e6ff93fddf67172c0ee26db74d236a17a9`.
- Independent certificate: `sha256:d80527e03481be10b16d4765742d717ebcefaa9f1830e2c1c37d6e40869a62e9`.
- External validation: `sha256:39850e79359dbd3ae8e41fe032474bcf197212f6ac0214cdfd90605919561761`.
- Engine receipt: `sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-PROBABILITY-STATISTICS-001-2a86d644881cdacb.json`.

14. Derivation 9: Exact finite optimization and retained optima

Claim identity: `SFT-MATH-OPTIMIZATION-001`

14.1 Question and necessity

Search and decision require lawful selection among alternatives. Exact dominance and tie retention prevent a familiar score, tolerance or traversal order from selecting the result.

The exact theorem statement is:

> *An admitted optimization problem is a complete canonical candidate carrier, a source-bound exact feasibility relation and a witnessed preference relation. Solutions are the complete undominated held selection; uniqueness is claimed only for a singleton, while ties and incomparable Pareto alternatives remain explicit. Constraints, decomposition and approximation are exact trace relations without floating objectives, negative penalties or free tuning parameters.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`.

14.2 Generated grammar and exact boundary

Orders supply witnessed preference and incomparability; graphs supply search relations and decomposable interfaces; exact arithmetic supplies part bounds. Nine choices force the complete undominated kernel.

Generation rule: Generate the complete product of candidate coverage, feasibility, preference, optimum selection, tie retention, constraint provenance, decomposition, approximation and extra-objective status.

Exact grammar boundary: All finite optimization instances generated from canonical candidates, exact predicates, held preference relations, complete dominance checks, interface-preserving decomposition and exact positive part bounds.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:a15ea2f582138e72734b4421a5d69ae92089f444f70597f7539532587a30ddbe`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-OPTIMIZATION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`candidates`	`sampled-candidates`	Sampling cannot prove that a better generated candidate was not omitted.	`complete-canonical-candidates` - Every registered candidate form occurs once.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`feasibility`	`implicit-constraint`	An implicit constraint has no candidate-level check trace.	`source-bound-verdicts` - Every candidate receives one exact verdict from each registered constraint.
`preference`	`floating-score`	A floating score imports scale and precision choices.	`witnessed-order-relation` - Every preference is an exact retained candidate pair.
`solution`	`first-found`	First-found depends on traversal presentation.	`complete-undominated-selection` - Every feasible candidate is compared and all undominated forms are retained.
`ties`	`silent-tie-break`	A silent tie-break adds a free preference.	`retained-equivalence-class` - All jointly optimal or incomparable surviving forms remain explicit.
`constraints`	`penalty-parameter`	A tunable penalty is a free parameter and can select the answer.	`exact-membership-predicate` - Each constraint is a source-bound exact allowed-form relation.
`decomposition`	`assumed-separability`	Shared constraints can invalidate separate optima.	`exact-interface-factorization` - Subproblems compose only when all shared interface relations are retained.
`approximation`	`decimal-error`	A decimal error imports rounding and a scale.	`exact-positive-part-bound` - The candidate/optimum relation is a generated exact part certificate.
`addition`	`extra-objective-or-tie-rule`	An added score or tie rule is not forced by the instance.	`no-extra-objective` - Only registered feasibility and preference relations select solutions.

14.3 Unique survivor, minimality and laws

> *The optimization kernel is complete canonical candidates, source-bound feasibility, exact preference, complete undominated selection, retained ties, exact constraints and interfaces, and positive-part bounds without an added objective.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- optimization may eliminate a candidate only with a retained feasible dominator witness.
- all undominated candidates survive and uniqueness is a separately checked singleton property.
- Pareto comparison retains incomparability across exact criteria.
- decomposition is lawful only with complete shared-interface factorization.
- approximation is an exact generated part relation rather than a rounded scalar.

14.4 Operational witnesses and unfavorable checks

- `feasible-selection` - The complete verdict relation retains exactly a and b. Result: PASS.
- `unique-optimum` - A witnessed a-over-b relation leaves a as the singleton optimum. Result: PASS.
- `ties-retained` - With no forced preference both feasible forms survive explicitly. Result: PASS.
- `constraint-membership` - The witness candidate is checked against every exact allowed set. Result: PASS.
- `pareto-incomparability` - Opposed exact criteria retain both incomparable forms. Result: PASS.
- `exact-bound` - Three generated cells against a two-cell optimum produce exact two-over-three. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling cannot prove that a better generated candidate was not omitted. Result: PASS; receipt `sha256:4fca7bb33759009f032e930b863d7c4fce12d4ce19f11686801c02efdbba245f`.

- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no floating objective or tolerance; no negative penalty term; no tunable regularizer or tie-break parameter; no unenumerated candidate domain Result: PASS; receipt `sha256:40faa5eb68ab3bcadce5ee1937d3b89e679a8f1fa2ba12fca91be6318f92420c`.

14.5 Depth-independent closure

Base: A One-candidate feasible carrier has that candidate as its unique complete undominated selection.

Successor: Adding one fresh candidate generates its feasibility verdicts and every preference pair with prior feasible forms. It is either eliminated by a retained dominator, eliminates witnessed dominated forms, or joins the retained optimal class; no prior optimum is silently discarded.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

14.6 Meaning and scientific consequence

Optimization is forced as elimination by exact witness rather than maximization of an imported floating score. Every candidate and feasibility verdict is present. A candidate is removed only when a retained feasible dominator exists. The complete undominated selection is the solution; singleton status alone licenses uniqueness. Multiple optimal or Pareto-incomparable forms remain visible rather than being broken by an undeclared convention.

14.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: feasible set, objective, optimum, Pareto frontier, constraint, approximation ratio. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

14.8 Exact exclusions and limitations

- no floating objective or tolerance.
- no negative penalty term.
- no tunable regularizer or tie-break parameter.
- no unenumerated candidate domain.

This closes exact finite optimization structure. Complexity of finding the complete frontier and laws for particular objective families belong to the algorithms and complexity branches.

14.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:435743b4bc243523b8c405514232bdc88a02fd48764040fa26c914085ae0f813`.
- Derivation seal: `sha256:11d1d87d66d732bca3efc296de283e8e5e11ed9d291dd083c10f86d6801277e9`.
- Independent implementation: `sha256:19c3f41f797dbf136a28392518b1ef3ff431214a314099a6080c8c5ea04edd95`.
- Independent certificate: `sha256:9cb0790543dac4fb7c8ff8bce76f783ccb1c96e84176f8cf59c203060920bfa4`.

- External validation: `sha256:7f6345c8109261bf2f63108a2613767ff27fae5f9d411bace11a33cc9046ddb3`.
- Engine receipt: `sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-OPTIMIZATION-001-3385414ceea0fd42.json`.

15. Derivation 10: Exact generated finite dynamical systems

Claim identity: `SFT-MATH-DYNAMICAL-SYSTEMS-001`

15.1 Question and necessity

Computation and natural modelling both require change through state. Exact transitions expose the laws of time, recurrence and information loss without importing continuous equations or stochastic causes.

The exact theorem statement is:

> *An admitted dynamical system is a complete canonical state carrier with a held transition relation. A trajectory retains every state and transition; time is the positive transition trace and the identity trajectory is empty One. Fixed forms, returns, recurrence, basins, reversibility and stability are admitted only through exhaustive generated witnesses. Merged predecessors remain recoverable only when their exact records are retained.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`.

15.2 Generated grammar and exact boundary

Graph paths supply evolution; optimization supplies complete invariant and basin selection; canonical forms supply state identity. Ten choices force the finite transition-and-record dynamics kernel.

Generation rule: Generate the complete product of state coverage, transition provenance, trajectory trace, time, fixed forms, return, basin/recurrence, reversibility, stability and extra-dynamic status.

Exact grammar boundary: All finite dynamics generated from canonical state carriers, held transition relations, complete paths, return traces, reachability basins, retained predecessor records and registered finite perturbation classes.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:4a398993d220a557338f9d3f2e0592838df6bccbb7f1d20274106c1a406ad684`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-DYNAMICAL-SYSTEMS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`states`	`partial-or-aliased-state`	Omitted or aliased states change possible evolution.	`complete-canonical-states` - Every generated state occurs once with canonical identity.
`transitions`	`opaque-update-rule`	An opaque update hides which relation cells exist.	`held-state-relation` - Every allowed state transition is a retained generated pair.
`trajectory`	`endpoint-only`	Endpoints erase intervening states and cannot prove reachability.	`complete-transition-trace` - Every adjacent transition is retained in order.
`time`	`continuous-clock-parameter`	A free real clock imports scale and continuum.	`positive-transition-count` - Elapsed time is the exact trace of generated transitions; identity is empty One.
`fixed`	`visual-sameness`	Presentation similarity does not prove dynamical identity.	`retained-identity-transition` - A fixed form has an explicit self relation.
`return`	`periodic-assertion`	A period name without a trace omits return structure.	`explicit-first-return-trace` - A retained nonempty path returns to a prior canonical state.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`basin`	`sampled-neighborhood`	Sampling cannot establish reachability for the registered perturbation class.	`complete-reachability-selection` - Every registered state is exhaustively tested for reachability to the invariant form.
`reversibility`	`infer-unrecorded-predecessor`	A merged image does not identify which predecessor occurred.	`retained-predecessor-record` - The exact source-to-image distinction is held alongside the step.
`stability`	`epsilon-parameter`	A chosen epsilon imports a free scale.	`registered-perturbation-closure` - Every explicitly generated perturbation form satisfies the invariant reachability condition.
`addition`	`extra-force-or-random-law`	An added force, differential equation or random update is not supplied by the state relation.	`no-extra-dynamic-law` - All properties follow from exact finite transition structure.

15.3 Unique survivor, minimality and laws

> *The dynamical kernel is complete canonical states, held transitions, complete trajectories, transition-count time, witnessed fixed/return/basin structure, retained-record reversibility and registered perturbation stability.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- trajectory composition is lawful exactly at an equal canonical interface state.
- time is a positive transition trace and identity duration is empty One.
- fixed, cyclic and recurrent behavior require explicit transition witnesses.
- many-to-one evolution loses predecessor distinction unless a source record is retained.
- stability is quantified only over a complete registered finite perturbation class.

15.4 Operational witnesses and unfavorable checks

- `well-formed` - Every witness transition remains in the complete state carrier. Result: PASS.
- `trajectory` - The a-to-b-to-terminal trace retains both allowed transitions. Result: PASS.
- `time` - Two transitions are retained exactly and an identity trajectory returns empty One. Result: PASS.
- `fixed-form` - The terminal witness has an explicit identity transition. Result: PASS.
- `first-return` - A repeated canonical state yields its exact first return trace. Result: PASS.
- `predecessor-retention` - Both merged predecessors remain distinct and an exact retained ledger reverses them. Result: PASS.
- `registered-stability` - Every registered witness perturbation reaches terminal. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omitted or aliased states change possible evolution. Result: PASS; receipt `sha256:644913b363fe507365e0c82e18a6847f33d91874acd59dea32a28f49e1ed33b4`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no continuous real time premise; no differential equation imported as a law; no stochastic transition cause; no unrecorded predecessor

reconstruction Result: PASS; receipt
`sha256:5c5f8dd163823e87bd822f6397adc6f8723ee60d3aa2defdfb5cc33a3ff76649`.

15.5 Depth-independent closure

Base: One state supplies its identity trajectory; no transition time or predecessor ambiguity is introduced.

Successor: Adding one transition appends one exact state pair to every extended trajectory. Fixed, return, basin, predecessor and stability obligations change only for paths that use the new relation and are rechecked completely.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

15.6 Meaning and scientific consequence

Dynamics is state-transition structure before any differential equation. Time is the exact positive trace of transitions, with the identity trajectory represented structurally by empty One. Fixed forms and cycles require explicit returns. Basins and finite stability classes are exhaustive reachability selections. If distinct predecessors merge, reversal requires the precise predecessor record; the image alone cannot recreate a distinction it does not contain.

15.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: state space, trajectory, fixed point, cycle, attractor, reversibility, stability. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

15.8 Exact exclusions and limitations

- no continuous real time premise.
- no differential equation imported as a law.
- no stochastic transition cause.
- no unrecorded predecessor reconstruction.

This closes exact finite dynamical systems. Continuum differential models may later appear only as measured or finite approximation correspondences, never as premises of this theorem.

15.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
  `sha256:366c10101f60725e0e8bc8f29c974d4a7707b564125c8b7608d655e2092ad5a2`.
- Derivation seal: `sha256:972a41752e953bd0b4d12f5c3c7d8e68c2aa2107a5aacaf9e330a2278b1a3a51`.
- Independent implementation:
  `sha256:6c685056fb0e01ef2133db2ab947796fbade3bd1e246c56479bbac46055751c0`.
- Independent certificate:
  `sha256:c8e806f9f4e6b74927f0bfe79e2c5b4351113323176e59ecde786d4121290172`.
- External validation:
  `sha256:f93adcb26bb1c7baeec979059b40325d370b5f2f3d5f50cd6885a5159b10362f`.
- Engine receipt: `sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe`.
- Engine receipt path:
  `receipts/engine/model_admitted/SFT-MATH-DYNAMICAL-SYSTEMS-001-3e0e44b65fbe660f.json`.

16. Derivation 11: Exact finite logic, inference and proof boundaries

Claim identity: `SFT-MATH-LOGIC-PROOF-001`

16.1 Question and necessity

Every branch depends on distinguishable valid and invalid derivations. The proof law makes admission mechanical while preventing conventional logical axioms or institutional authority from entering silently.

The exact theorem statement is:

> *An admitted proposition is a canonical distinguished form with a held orientation; denial is the retained complementary orientation, not a negative number. Conjunction retains joint proofs, disjunction retains a chosen generated branch and its alternatives, implication is a registered premise-to-conclusion transition, and a proof is a complete dependency trace of accepted rules. Consistency, soundness and completeness are machine-checkable only relative to the exact registered finite grammar and rule set.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-ORDER-LATTICE-001`.

16.2 Generated grammar and exact boundary

Canonical forms supply proposition identity; held fibres supply opposed orientations; discrete relations supply rules; order supplies proof-strength relations. Ten choices force the replayable finite proof kernel.

Generation rule: Generate the complete product of proposition identity, denial, conjunction, disjunction, implication, inference provenance, consistency, soundness, grammar completeness and extra-logical status.

Exact grammar boundary: All propositions and finite proofs generated from canonical forms, two held orientations, joint and alternative constructions, registered inference transitions and complete dependency traces.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:c3171696c5a96a3fa85bb809773cca0fc438e1d7196b6657aa3f94ce06e1e622`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-LOGIC-PROOF-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`proposition`	`presentation-sentence`	A sentence presentation can alias or hide the judged form.	`canonical-distinguished-form` - The exact generated form is the proposition identity.
`denial`	`negative-truth-value`	A negative magnitude is outside the admitted domain.	`complementary-held-orientation` - Denial retains the same form with the other exact fibre orientation.
`conjunction`	`single-side-proof`	One side cannot establish the joint claim.	`joint-complete-proof` - Both constituent proof traces are retained together.
`disjunction`	`unidentified-choice`	An unidentified choice erases which alternative is supported.	`held-alternative-with-support` - The held branch and complete generated alternative family are retained.
`implication`	`truth-table-import`	A borrowed table installs conventional logical values.	`proof-carrying-transition` - A registered rule maps exact premises to an exact conclusion.
`inference`	`conclusion-only`	A bare conclusion erases its premises and rules.	`complete-dependency-trace` - Every step retains available premises, rule identity and conclusion.
`consistency`	`assumed-consistent`	An assertion does not search for opposed admitted orientations.	`no-opposed-admission` - No form and its complementary orientation are jointly admitted.
`soundness`	`authority-or-name`	Authority cannot replace validation against registered rules.	`rule-and-witness-preservation` - Every accepted step matches a registered source-bound rule and preserves its witnesses.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`completeness`	`unbounded-global-claim`	A finite proof kernel cannot establish an ungenerated universal domain.	`registered-grammar-exhaustion` - Every proposition in the exact generated grammar has its declared decision evidence.
`addition`	`extra-logical-axiom`	An extra axiom is not supplied by the canonical form and rule traces.	`no-extra-logical-axiom` - All admitted inference is registered and witnessed.

16.3 Unique survivor, minimality and laws

> *The logic/proof kernel is canonical distinguished propositions, complementary held denial, joint and held alternative proof forms, proof-carrying implication, complete inference traces, explicit consistency and soundness, and completeness only over the exhausted registered grammar.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- denial changes held orientation while preserving proposition identity.
- conjunction and disjunction retain their complete source proof structure.
- an implication is admitted through a registered proof-carrying transition.
- a valid proof is replayable step by step from its premises and rule registry.
- completeness is always indexed by an explicit generated grammar boundary.

16.4 Operational witnesses and unfavorable checks

- `held-denial` - Denial preserves P while changing only its held orientation. Result: PASS.
- `joint-proof` - Conjunction retains both exact constituent proofs. Result: PASS.
- `held-disjunction` - Disjunction retains the supported branch and both alternatives. Result: PASS.
- `proof-replay` - The registered P-to-Q step replays from its available premise. Result: PASS.
- `tampered-rule-control` - Changing the rule identity invalidates the same conclusion trace. Result: PASS.
- `consistency` - P and Q are consistent, while P and its held denial are not. Result: PASS.
- `bounded-completeness` - Both registered proposition forms have explicit held decisions. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A sentence presentation can alias or hide the judged form. Result: PASS; receipt `sha256:6d035b68865a940e61e80880013f484689afa3fbfab68a65b3604a51f203a98c`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported truth-value arithmetic; no negative truth magnitude; no authority-based proof admission; no global completeness beyond a generated grammar Result: PASS; receipt `sha256:5dff4ae767ffc5d54e6bf8debb3ec71db9285d631376f5ba4b148be47b1cda93`.

16.5 Depth-independent closure

Base: One proposition form and its two orientations provide the first decidable grammar cell.

Successor: Adding one proposition or rule generates its complementary orientation, joint and alternative combinations, and all new premise tuples. Consistency and proof replay are rechecked for precisely those new forms and steps.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

16.6 Meaning and scientific consequence

Logic is reconstructed as distinguished canonical forms and proof-carrying transitions. Denial changes a held orientation; it is not negative truth magnitude. Conjunction retains both supporting traces and disjunction retains the supported alternative and the full alternative family. A proof is valid only by stepwise replay against registered rules. Soundness and completeness are therefore exact but explicitly relative to the generated grammar; no unbounded global completeness claim is smuggled into the result.

16.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: proposition, negation, conjunction, disjunction, implication, proof, soundness, completeness. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

16.8 Exact exclusions and limitations

- no imported truth-value arithmetic.
- no negative truth magnitude.
- no authority-based proof admission.
- no global completeness beyond a generated grammar.

This closes finite generated proof theory. Computability, self-reference, undecidability and incompleteness boundaries require the later formal-computation branch and are not claimed here.

16.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:5b549044ca00a139f499155286a06ed09af0735799cf371650de8fd8d336a66d`.
- Derivation seal: `sha256:849294dd48e23a2aefd379804094c01132234adad33b937a3164b3c73f8230bf`.
- Independent implementation: `sha256:636f912fdacfb994061d4018dcbfa82b20c5f99693af0f3f5e9be0ee84b13a42`.
- Independent certificate: `sha256:dfeb338769fcf4b551277891fd7c5576f15ce79e533addbb15e6e4eed87a288a`.
- External validation: `sha256:d3cc342c7a7bf4b50450ae303b15dd250328f761236f869150456ba1550c3cb9`.
- Engine receipt: `sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-LOGIC-PROOF-001-3d366d57b26d002c.json`.

17. Derivation 12: Forced category, type and compositional structure

Claim identity: `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`

17.1 Question and necessity

Every later science must compose local derivations without losing interfaces or provenance. The forced path kernel supplies this language while keeping category and type concepts downstream of Fold structure.

The exact theorem statement is:

> *Canonical Fold forms act as objects and proof-carrying transitions as arrows. Identity is the empty-One return, composition is forced exactly by equal interfaces, and association follows from canonical complete path flattening. Types are source-bound predicates over generated carriers; products are complete joint pair support and sums retain held alternative labels. Functors preserve interfaces, identity and composition, while naturality is equality of the two complete composite traces.*

Admitted dependencies: `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ALGEBRA-001`, `SFT-MATH-LOGIC-PROOF-001`.

17.2 Generated grammar and exact boundary

Graphs supply paths, algebra supplies operation association, and logic supplies proof-carrying transitions. Ten choices force exact interfaces, identities, flat composition, types and preservation structures.

Generation rule: Generate the complete product of objects, arrows, identity, composition, association, types, product/sum structure, functoriality, naturality and extra-composition status.

Exact grammar boundary: All finite compositional structures generated from canonical objects, proof-carrying arrows, exact interface matching, empty-One returns, flattened paths, source-bound type predicates and held joint/alternative forms.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:61559f26b9f1269604ea49148c48493a1565fa04b140255215af667d0a20aa20`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-CATEGORY-TYPE-COMPOSITION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`objects`	`presentation-objects`	Presentation names can alias distinct structures.	`canonical-Fold-objects` - Every object is an exact canonical generated form.
`arrows`	`endpoint-only-arrow`	Endpoints alone erase the transformation proof.	`proof-carrying-transition` - An arrow retains source, target and complete transition trace.
`identity`	`assumed-neutral-map`	A named neutral map has no return witness.	`empty-One-return` - The zero-step structural return preserves the same object without numerical zero.
`composition`	`presentation-concatenation`	Textual concatenation can join unequal interfaces.	`exact-interface-match` - The first target and second source must be the same canonical object.
`association`	`borrowed-category-axiom`	Importing an axiom would reverse the derivation order.	`canonical-path-flattening` - Both groupings retain the same ordered elementary transition trace.
`types`	`informal-class-name`	A class name need not decide carrier membership.	`source-bound-form-predicate` - Every generated carrier form receives one exact membership verdict.
`constructors`	`borrowed-set-constructor`	A borrowed constructor imports its ambient universe.	`joint-pairs-and-held-alternatives` - Products are complete pair support; sums retain left/right held provenance.
`functor`	`arbitrary-renaming`	Renaming can break interfaces and composites.	`identity-and-composition-preservation` - Object and arrow maps retain sources, targets, returns and composites.
`naturality`	`diagram-by-appearance`	A drawn square cannot prove both routes equal.	`equal-complete-paths` - Both interface-compatible composite routes have the same canonical trace.
`addition`	`extra-composition-axiom`	An extra universal property is not supplied by the registered traces.	`no-extra-composition-axiom` - Only generated interfaces, paths and preservation witnesses are admitted.

17.3 Unique survivor, minimality and laws

> *The compositional kernel is canonical Fold objects, proof-carrying arrows, empty-One identity, exact-interface composition, path-flattening association, source-bound types, joint products, held sums, preserving functors and equal-path naturality.*

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

- identity preserves source and target and contributes no nonidentity transition.
- composition exists exactly at a canonical interface match.
- associativity is equality of flattened complete transition traces.
- products preserve both coordinates and sums preserve the held source alternative.
- functoriality and naturality are finite replayable preservation witnesses.

17.4 Operational witnesses and unfavorable checks

- `interface-composition` - f then g composes from A to C with both traces retained. Result: PASS.
- `identity` - Empty-One return preserves the interface and adds no elementary transition. Result: PASS.
- `association` - Both groupings of f, g and h flatten to the same complete path. Result: PASS.
- `type-predicate` - The witness predicate retains exactly its admitted carrier forms. Result: PASS.
- `product-sum` - Joint products and held sums retain both source coordinates. Result: PASS.
- `functor-interface` - The identity transport preserves the witness arrow interface. Result: PASS.
- `naturality-path` - Two independently assembled routes with the same elementary trace commute. Result: PASS.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Presentation names can alias distinct structures. Result: PASS; receipt `sha256:0b1412200b24f35693db81fac968726a12b5f6d4a14d7866a33dfdfa597ffe8a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported category axioms; no ungenerated universe of all objects; no impredicative type constructor; no universal property without an exhaustive finite witness Result: PASS; receipt `sha256:b3ab5d7231f463ee56b4759ebf5e5a9ec0c2a2f15ff312709eed57a7ba33281d`.

17.5 Depth-independent closure

Base: One canonical object supplies its empty-One identity arrow and the first singleton type.

Successor: Adding one object or arrow generates its identities, every exact interface-compatible composite, product and sum cells, type verdicts and preservation squares. Closure follows by appending its trace to prior flat paths.

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

17.6 Meaning and scientific consequence

Composition is the final mathematical unifier of the branch. Canonical forms are objects and complete proof-carrying transitions are arrows. Equal interfaces force lawful composition; empty-One return supplies identity; and associativity is equality of flattened elementary paths. Types are exact predicates over a registered carrier. Products retain joint coordinates, sums retain held origin, and functorial or natural structure is admitted by replayable path-preservation witnesses.

17.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: category, object, morphism, identity, composition, type, product, sum, functor, natural transformation. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

17.8 Exact exclusions and limitations

- no imported category axioms.
- no ungenerated universe of all objects.
- no impredicative type constructor.
- no universal property without an exhaustive finite witness.

This closes the finite compositional kernel. Any stronger universal construction must be separately generated and exhaustively witness its claimed mapping property inside an explicit grammar.

17.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:8962e0d82e8291e69ae4b1477518316402aea5b3e663f08b418e88bec287d135`.
- Derivation seal: `sha256:756d7d6dce776fc1fb4b3906e2203e4e2da4e1ded6cf733eb041334eb7cb5c02`.
- Independent implementation: `sha256:ced0eda6a320565b205b93618d604af7b494f3ad001044d71c87df0f4c281f90`.
- Independent certificate: `sha256:1d380a6b0d0ce9d32cfdff52fae121315257e3e561da9379ab5509c835680dad`.
- External validation: `sha256:fc7b4ce671b53f4d62da07705c4fc5d92b0195d5218a30f2a40a773a2fb7c981`.
- Engine receipt: `sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-CATEGORY-TYPE-COMPOSITION-001-47873a83a90e29fa.json`.

18. Cross-derivation synthesis

The twelve theorems form one dependency chain rather than a list of renamed fields. Exact arithmetic gives lawful trace junction, pair-cell product, refinement and comparison. Discrete mathematics turns those operations into canonical carriers, selections, relations, maps and induction. Combinatorics then generates complete families of choices; graph theory holds a relation from complete pair support and promotes it to path, cycle, cut and network structure. Algebra asks which complete operation relations close, associate, return and map. Order retains only witnessed comparison and derives conditional extremal operations.

Geometry and topology use graph paths, incidence and order closure to derive the finite structures computation actually requires. Probability uses complete combinatorial support and exact parts to quantify observation classes without altering deterministic state. Optimization uses order to preserve feasible undominated alternatives. Dynamics uses graph transitions to define trajectory, return, basin and record-dependent reversal. Logic then turns distinguished forms and relations into proof-carrying inference. Category, type and composition close the chain by showing how exact objects and transformations compose without losing interface provenance.

No result licenses backward importation. Because category-like composition is derived last, it cannot be an axiom used to select arithmetic. Because probability is downstream of complete deterministic support, it cannot install an ontic random cause in the earlier state law. Because geometry is finite incidence before continuum correspondence, an irrational coordinate cannot retroactively become foundational proof evidence.

19. Exact arithmetic without zero, negatives or irrational proof values

The prohibition on semantic numerical zero does not make empty, identity or no-transition cases inexpressible. It forces them to retain their structural origin. Empty selection is empty One; identical traces return same-form; an identity arrow is the empty-One return; identity duration is the one-state trajectory with no transition. These forms are typed structural records, not a scalar that can be silently substituted across domains.

Negative quantities are likewise unnecessary for exact comparison. If two traces pair completely they are the same form. Otherwise the unmatched positive occurrences are retained and a held label records which trace owns them. Network balance pairs positive token identities. Logical denial changes held orientation. Algebraic return uses a complementary carrier mate. No proof relies on subtracting a larger magnitude from a smaller one.

Irrational, imaginary and floating values are excluded from formal admission because the generated Foundation supplies exact positive parts and finite refinement, not an ungenerated completed field. This does not deny that conventional sciences record such descriptions. It establishes that they may enter only as correspondence, finite approximation or measured values after the law is sealed. A later approximation theorem must state its generated sequence, exact error relation, convergence boundary and measurement custody.

20. Probability, statistics and superdeterminism

The probability theorem is often the point at which a deterministic model is tempted to import an incompatible prior. SFT does not do so. A microstate is one exact generated form. The registered support contains every such form once. Observation is a total classification relation from microstates to retained observation labels. States with the same image form an observation class. Uncertainty is precisely the collection of distinctions present in support but closed by that observation; the underlying transition law remains deterministic.

An event is a held support selection. For a nonempty event, probability quantity is the exact positive count of held states relative to the exact positive count of the complete support. The empty event remains empty One and never passes through numerical-zero arithmetic. Conditioning restricts both event and whole to a nonempty held condition before forming the exact part. Independence is not lack of a fitted correlation; it is the structural fact that joint support is the complete pair-cell product and the event pairing factorizes.

Statistical summaries are allowed because they are transparent functions of complete finite records. Ties are retained. A natural-data claim would require a separate empirical registration, frozen source and target custody. The target must remain closed until after the derivation seal; all rows, including failures, must be preserved. No natural dataset was needed or used to admit the formal probability theorem in this paper.

21. Formal proof versus empirical validation

These twelve claims are formal structural theorems. Their empirical content is computational execution: the candidate products are actually generated, decisions actually run, controls are deliberately perturbed, separate validator processes execute and receipts are independently replayed. This execution evidence demonstrates that the declared finite grammars and certificates have the reported machine behavior. It is not observational evidence about an external natural system, and the paper does not mislabel it as such.

When a later branch asserts a relation to natural data, the Foundation measurement boundary applies. Formal law selection and target evaluation are different phases with different custody. A blind opaque predictor cannot replace a derivation chain because a score alone does not identify premises, alternatives, eliminations, source identity, failure conditions or whether the target influenced the law. An empirical SFT claim must expose all of those objects and halt if custody or capability isolation fails.

22. Machine implementation and complete reproduction

The definitive cross-platform logical command is `sft verify-all`, invoked with the host's standard Python launcher:

```
python3 -m sft verify-all   # macOS and most Linux systems
py -m sft verify-all        # standard Windows Python launcher
```

No Docker image, virtual machine, network service or third-party Python package is required for the scientific verification path. The command validates repository structure, runs all unit, unfavorable-control and end-to-end tests under line coverage, requires 100 percent executable-line coverage of the 15-module admission engine, loads the ordered execution manifest, rebuilds each source manifest, reruns every candidate and control, launches every independent validator and compares each new receipt byte-for-structure with the admitted stored receipt.

The verified local report for this paper is:

```
SFT COMPLETE VERIFICATION: PASS
unit and end-to-end tests passed: 111
core engine executable-line coverage: 1264/1264 (100%)
core engine modules covered: 15
registered derivations independently rerun: 22
```

The coverage statement concerns the engine implementation; mathematical closure comes from the claim-specific candidate grammar, decision vector, unique survivor, minimality, named-shape uniqueness and induction certificate. Neither substitutes for the other.

23. Independent criticism and invalidation routes

A reviewer need not accept the vocabulary or conclusions to reproduce the result. The strongest direct criticism is to identify a candidate inside a declared boundary that the registered product omits. That attacks completeness. A second route is to show that a rejected coordinate actually preserves every requirement, defeating minimality, or that another product member survives, defeating uniqueness. A third is to invalidate the base or successor certificate. A fourth is to provide an input inside the declared domain that violates an operational law.

The repository also supports mechanical attacks: alter a source after registration, change a candidate identity, remove a decision, add a survivor, fail a control, tamper with a certificate, reorder the execution manifest or change a receipt. The engine halts at the exact violated gate. Hashes do not establish truth; they establish which exact sources and artifacts were checked so an altered object cannot silently inherit a prior status.

24. Consequences for the remaining knowledge tree

The completed branch supplies the exact resources the next branch may cite: symbols can be canonical distinguished forms; encodings can be complete maps; information quantity can be an exact relation between support distinctions and observation; channels can be relations; entropy and uncertainty must respect deterministic support and exact parts; coding can use combinatorial word families, graph paths and algebraic operations. None of those downstream laws is admitted by this paper merely because its mathematical vocabulary now exists.

Formal computation may build states from canonical forms, transitions from relations, machines from dynamical systems, resource orders from exact positive traces, semantics from proof-carrying composition and quantum support from complete Fold word families. Physics and later natural sciences may use finite geometry, dynamics, exact probability and optimization, but must separately derive their laws and pass blind empirical validation where observation is required. Applications remain frontier translations until the relevant science branches close.

25. Scope, limitations and next branch

Closure is exact at the frozen generated-finite Mathematics boundary. It does not assert a completed universe of all sets, a real or complex continuum, an infinite-dimensional space or every named theorem of conventional mathematics. It derives the general structural kernels needed to generate and test finite exact instances. Named special structures must still expose their carrier and property evidence. Continuum expressions can enter only through separately registered finite approximation or measurement claims.

No official natural empirical claim is made in this branch. No application experiment has been used. The next dependency branch is Information Science: symbols and distinguishability, encoding and decoding, information quantity, entropy and uncertainty, compression, channels and capacity, noise and error, coding, mutual and conditional information, and classical-probabilistic-quantum information correspondence.

26. Open-science status, rights and participation

Credentials cannot rescue a failed gate, and lack of credentials cannot prevent a reproducible criticism from being evaluated. Independent replications, omissions, counterexamples and submissions are invited through Maria.Smith.Sftoe@gmail.com and https://discord.gg/ucwGryVxGr.

27. Conclusion

The Mathematics branch closes twelve dependency-ordered kernels through 7,424 generated alternatives, twelve unique survivors, twelve depth-independent certificates and twelve implementation-distinct validations. It reconstructs exact arithmetic, finite discrete and combinatorial structure, relations and graphs, witnessed algebra and order, computational geometry and topology, deterministic-support probability, optimization, dynamics, proof and composition without importing conventional answer-producing models.

The unification is methodological as well as mathematical: every object has a generated boundary, every elimination has a reason and proof identity, every closure has an induction certificate, every adverse control is preserved and every admitted result can be independently rerun. The branch is accepted within its exact boundary because that complete chain closes - not because a prior authority, application score or familiar notation selects it.

Appendix A. Authoritative receipt identities

Claim	Engine receipt
SFT-MATH-EXACT-ARITHMETIC-001	sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418
SFT-MATH-DISCRETE-001	sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb
SFT-MATH-COMBINATORICS-001	sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968
SFT-MATH-GRAPH-NETWORK-001	sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a
SFT-MATH-ALGEBRA-001	sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd
SFT-MATH-ORDER-LATTICE-001	sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6
SFT-MATH-GEOMETRY-TOPOLOGY-001	sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e
SFT-MATH-PROBABILITY-STATISTICS-001	sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b
SFT-MATH-OPTIMIZATION-001	sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849
SFT-MATH-DYNAMICAL-SYSTEMS-001	sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe
SFT-MATH-LOGIC-PROOF-001	sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813
SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8

Appendix B. Complete evidence route

For every claim identifier `<CLAIM>`, the complete review route is:

```
claims/<CLAIM>/registration.json
claims/<CLAIM>/WHY_DERIVATION_CHECK.md
```

```
claims/<CLAIM>/candidate_census.json
claims/<CLAIM>/elimination_receipt.json
claims/<CLAIM>/controls.json
claims/<CLAIM>/certificate.json
claims/<CLAIM>/execution.py
claims/<CLAIM>/independent_validator.py
receipts/engine/model_admitted/<CLAIM>-<receipt-prefix>.json
```

Executable law sources are under `sft/mathematics/`. `sft/mathematics/catalog.py` fixes dependency order. `census/claims.json` is the model-admitted projection; `census/execution_manifest.json` is the complete replay order; `publications/inventories/mathematics.json` freezes scope. The separate paper evidence map binds every derivation section to exact source, claim-package and receipt hashes.

Appendix C. Reproducibility interpretation

A successful rerun proves that the checked sources deterministically reproduce the registered censuses, decisions, closures, controls, independent certificates and receipts on the reviewing host. It does not compel agreement with the grammar boundary. Scientific review therefore has two complementary tasks: reproduce the artifact and scrutinize whether the declared grammar exhausts the stated boundary. The repository makes both tasks explicit.

References

1. Smith M. *From Nothing to Fold: A Premise-Free, Parameter-Free and Machine-Closed Foundation for Smithian Fold Theory*. Ernos Labs Foundation Branch Paper 001. 2026. doi:10.5281/zenodo.21515629.
2. Smith M. *Smithian Fold Theory Scientific Constitution*. V3 clean-room repository, `CONSTITUTION.md`, 2026.
3. Ernos Labs. *The Single SFT Admission Engine*. V3 clean-room repository, `docs/ENGINE_AUTHORITY.md`, 2026.
4. Ernos Labs. *Comprehensive Branch-Paper Protocol*. V3 clean-room repository, `publications/BRANCH_PAPER_PROTOCOL.md`, 2026.
5. Ernos Labs. *Mathematics Frozen Current-Knowledge Inventory*. V3 clean-room repository, `publications/inventories/mathematics.json`, 2026.